



ST BENEDICT'S

SUBJECT

**Physical Science
Paper 2**

DATE

22 November 2021

GRADE

11

MARKS

200

EXAMINER

Mrs Govender

MODERATOR

Mr du Toit
Mr McCoy

NAME

DURATION

3 hours

CLASS

LOW ORDER

55%

HIGH ORDER

45%

INSTRUCTIONS

1. This question paper consists of **16** pages, an Answer Sheet of 2 pages and a Data Sheet of 2 pages. Please make sure that your question paper is complete.
2. Answer ALL the questions.
3. Read the questions carefully.
4. Use the data and formulae whenever necessary.
5. **Start each question on a new page.**
6. Show your working in all calculations.
7. Units need not be included in the working of calculations, but appropriate units should be shown in all answers.
8. Express all answers to TWO decimal places unless told otherwise.
9. It is in your own interest to write legibly and to present your work neatly.

Question	1	2	3	4	5	6	7	8	9	Total
Total	20	23	12	25	17	18	29	24	32	200

QUESTION 1 – MULTIPLE CHOICE

20 MARKS

Answer the following multiple-choice questions on the separate ANSWER SHEET. Make a cross (X) in the box corresponding to the letter which you consider to be the most correct.

1.1 The chemical formula for sodium sulphite is: (2)

- A. NaSO_3
- B. Na_2SO_3
- C. NaSO_4
- D. Na_2S

1.2 Which of the following best describes the particles and how the electrons are involved in the process of chemical bonding? (2)

	Particles	Electrons
A.	Non-metal ions only	Transferred
B.	Metal and Non-metal ions	Transferred
C.	Metal and Non-metal ions	shared
D.	Non-metal ions only	shared

1.3 Which of the following substances will conduct electricity as a solid? (2)

- A. Diamond
- B. Graphite
- C. Dry ice
- D. NaCl

1.4 Which one of the following statements represents the best explanation for the term “heat-of-reaction”?

- A. Reactions which transform chemical potential energy into thermal energy.
- B. The net change of kinetic energy of the system.
- C. Reactions which transform thermal energy into chemical potential energy
- D. The net change of chemical potential energy of the system.

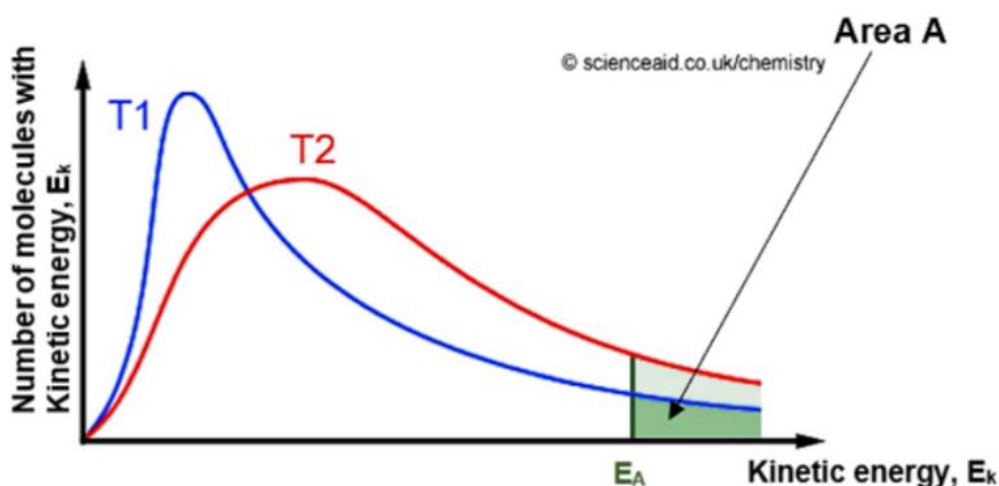
1.5 Which one of the following contains the most polar covalent bonds? (2)

- A. HF
- B. CH_4
- C. SO_2
- D. SiI_2

1.6 How many **atoms** are present in 1 mol of $(\text{NH}_4)_2\text{SO}_4$

- A. $6,02 \times 10^{23}$
- B. $5,42 \times 10^{24}$
- C. $9,03 \times 10^{24}$
- D. $8,43 \times 10^{24}$

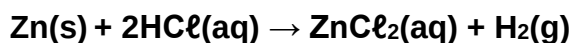
1.7 The diagram below shows the Maxwell-Boltzmann curves for the particles of a sample of a liquid at two different temperatures, T1 and T2. (2)



Which one of the following statements about this graph is **incorrect**?

- A. The temperature of T1 of the liquid is less than the temperature T2 of the liquid.
- B. The darkly shaded area A, represents the number of particles with enough of kinetic energy to react at T1.
- C. The average kinetic energy at T1 is greater than the average kinetic energy at T2.
- D. The rate of reaction at T1 is less than the rate of reaction at T2.

1.8 Which one of the options below is the best description of reaction rate for the following reaction? (2)

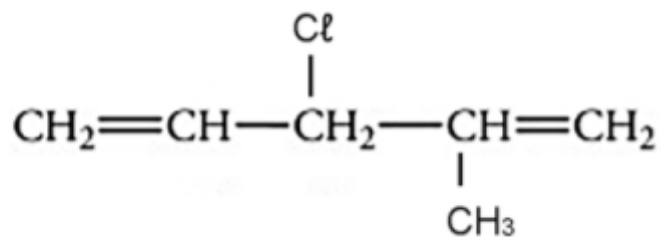


The reaction rate is...

- A. The time it takes to use up all the reactants.
- B. The decrease in concentration of the Zn per unit time.
- C. The time it takes for one of the reactants to be used up.
- D. The increase in concentration of the ZnCl_2 per unit time.

1.9 Choose the correct IUPAC name for the molecule shown below:

(2)



- A. 3-chloro-4-methylpenta-1,4-diene
- B. 2-methyl-3-chloro-1,4-pentene
- C. 3-chloro-2-methylpenta-1,4-diene
- D. 2-methyl-3-chloropenta-1,4-diene

1.10 Which one of the following reactions will NOT take place with an alkene?

(2)

- A. Hydration
 - B. Hydrogenation
 - C. Combustion
 - D. Dehydration
-

QUESTION 2

23 MARKS

2.1 Consider the table below showing the melting points of three solid oxide compounds.

	Substance	Melting point (°C)
A	Li_2O	1438
B	SnO_2	1630
C	SiO_2	1710

2.1.1 Explain why Li_2O does not conduct electricity as a solid. (2)

2.1.2 Lithium loses one electron to form a lithium ion when Li_2O forms.

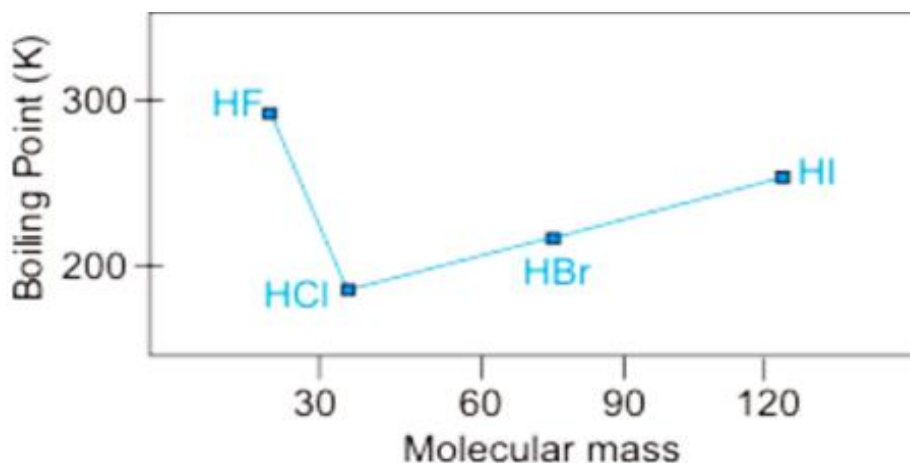
a) Name the process that takes place when lithium loses electrons. (1)

b) Write an equation to show how the process takes place. (2)

2.1.3 Name the intramolecular bond found in SiO_2 . (1)

2.1.4 Explain why SiO_2 has such a high melting point compared to the other SnO_2 and Li_2O . (3)

2.2 The graph below shows the trend in boiling points of hydrogen halides



2.2.1 Define an *intramolecular bond*. (2)

2.2.2 Use electronegativity differences to identify the type of intramolecular bonds present in HCl . (2)

2.2.3 Name the specific type of intermolecular force found in HF but not in HCl , HBr or HI . (1)

2.2.4 Explain the origin the intermolecular forces mentioned in 2.2.3. (3)

2.2.5 Draw the Lewis diagram for one molecule of HCl (2)

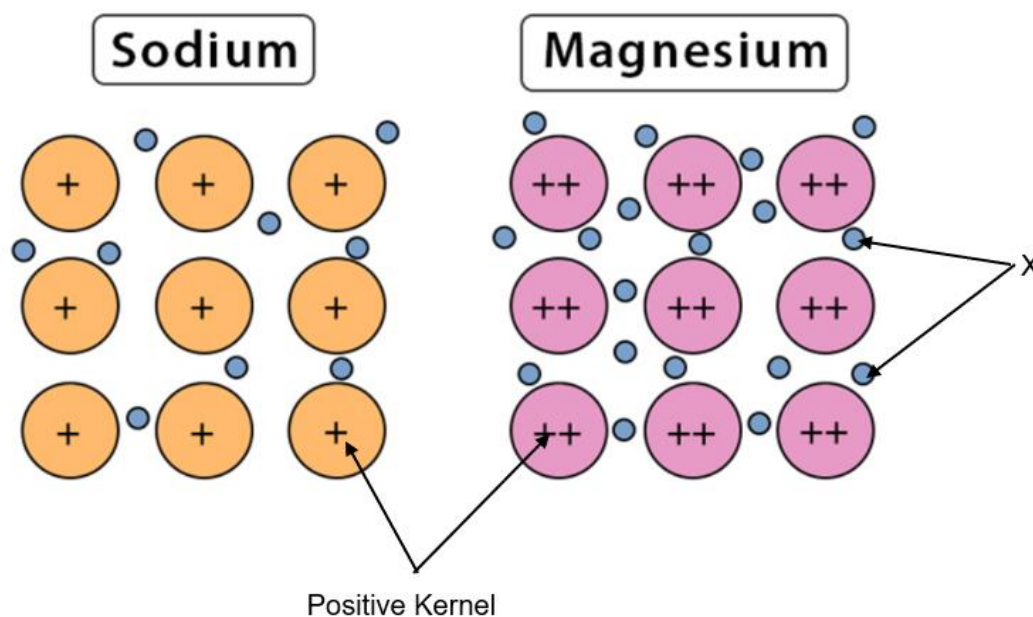
2.2.6 Indicate the polarity of the HCl molecule on your Lewis diagram by indicating the atom that is slightly + and slightly -. (1)

2.2.7 Explain the difference in boiling point between HBr and HI by referring to their intermolecular forces. (3)

QUESTION 3

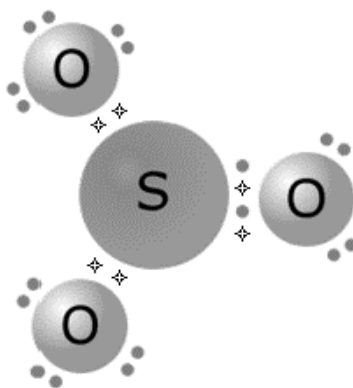
12 MARKS

3.1 The diagram below shows the metallic lattice for sodium and magnesium.



- 3.1.1 Define *metallic bonding*. (2)
- 3.1.2 Explain what is meant by a positive kernel. (2)
- 3.1.3 Provide a label for X. (1)
- 3.1.4 Use the diagrams to explain why the metallic bond in magnesium is stronger than the metallic bond in sodium. (2)

3.2 The diagram below shows dative of covalent bonding in sulphur trioxide.



- 3.2.1 Mention two requirements that must be met for this type of bonding to occur. (2)
- This question must be answered on the ANSWER SHEET.
- 3.2.2 Label the following bonds on the diagram: (3)
- a) dative bond
 - b) sigma bond
 - c) pi bond

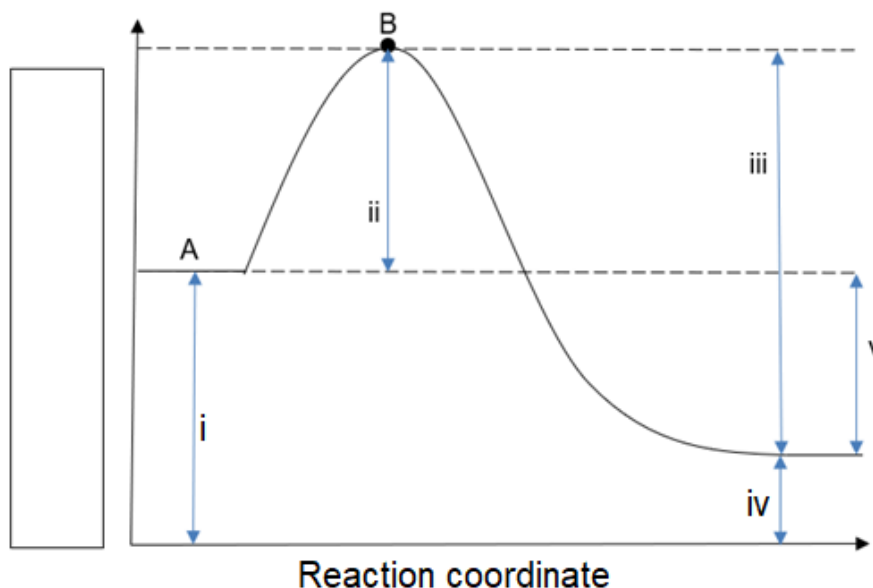
QUESTION 4

25 MARKS

In order to investigate the rate of the reaction between a carbonate and an acid, calcium carbonate and 2 mol.dm^{-3} hydrochloric acid react in a reaction vessel. The balanced chemical equation for this reaction is:

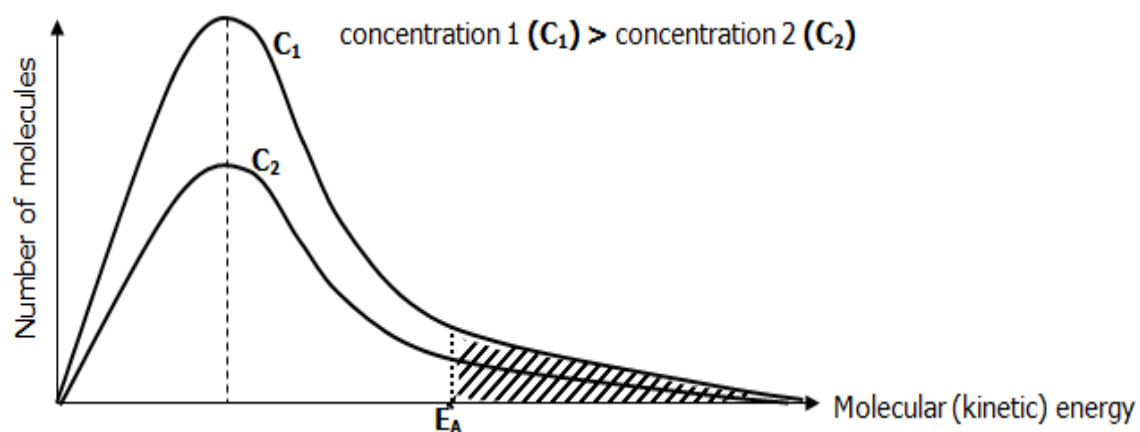


Consider the potential energy profile for this reaction as illustrated below:



- 4.1 Provide a label for the y axis. (2)
- 4.2 The graph has been labelled i – v to represent the various energies that are illustrated by this energy profile. Identify each of these energies on the profile. This question must be answered on the ANSWER SHEET provided. (5)
- 4.3 Using the molecular collision theory, explain why the chemical reaction must gain potential energy between position A and position B according to the energy profile. (2)
- 4.4 Provide a name for the position on the graph labelled B and what significant process takes place at this point. (2)
- 4.5 Does the graph represent an exothermic or an endothermic reaction? Explain how you made this conclusion. (3)
- 4.6 4.6.1 Define a *catalyst*. (2)
- 4.6.2 The same reaction takes place in the presence of a catalyst. How would the energy at point B change? State INCREASES, DECREASES or REMAINS THE SAME. (2)
- 4.6.3 How would the presence of a catalyst affect the stability of point B? Explain your answer. (3)

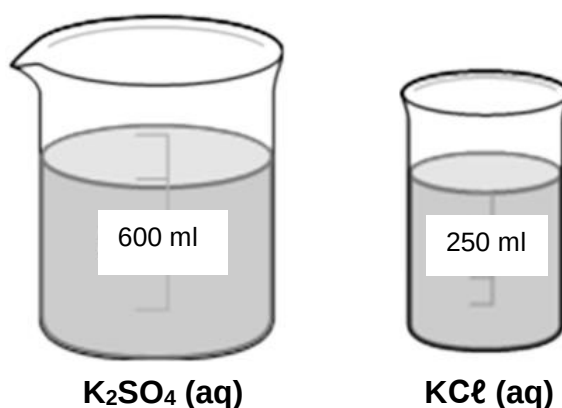
- 4.7 The concentration of hydrochloric acid (HCl) was increased. **Use the Maxwell-Boltzmann curve** below to explain how the increase in concentration of hydrochloric acid (HCl) affects the reaction rate. (4)



QUESTION 5

17 MARKS

- 5.1 A chemist makes a $0,5 \text{ mol}\cdot\text{dm}^{-3}$ standard solution of potassium sulphate($\text{K}_2\text{SO}_4(\text{aq})$) using the usual equipment.
- 5.1.1 Define a *standard solution*. (2)
- 5.1.2 Calculate the mass of K_2SO_4 needed to make 500 cm^3 of this standard solution. (4)
- 5.1.3 Should the solid K_2SO_4 be added to a volumetric flask before or after distilled water is added? Briefly explain your answer. (2)
- 5.2 In the diagram below, the concentration of the K_2SO_4 solution is $0,8 \text{ mol}\cdot\text{dm}^{-3}$ while the concentration of the KCl solution is $0,3 \text{ mol}\cdot\text{dm}^{-3}$.

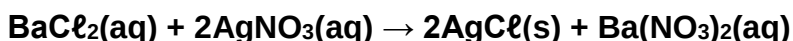


- 5.2.1 Write balanced dissociation equations for both solutions. No phase indicators are needed. (4)
- 5.2.2 The two solutions are mixed. What is the concentration of K^+ ions in the mixture? (5)
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QUESTION 6

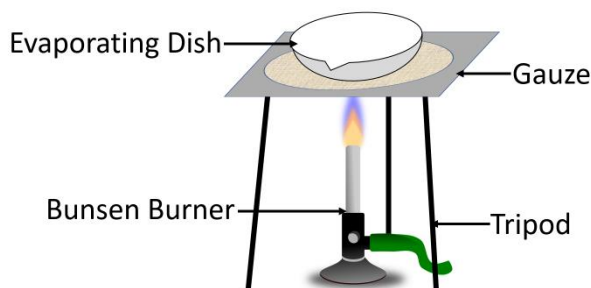
18 MARKS

- 6.1 Aidan and Ryan have 672 dm³ of ammonia, NH₃, and 1101,2 dm³ of carbon dioxide, CO₂, at STP. Aidan says that they have the same amount of substance, but Ryan says they don't. Who is correct? Explain your answer **without** using any calculations. (4)
- 6.2 Barium chloride and silver nitrate react according to the following balanced equation to produce silver chloride and barium nitrate.



If you have 1,39 g of BaCl₂ and 2,69 g of AgNO₃:

- 6.2.1 Prove that barium chloride (BaCl₂) is the limiting reagent. (4)
- 6.2.2 What mass of silver chloride (AgCl) will be produced (2)
- 6.2.3 Calculate the percentage yield if 1,80 g of AgCl is actually collected after the reaction is complete. Express your answer to the nearest WHOLE NUMBER. (4)
- 6.3 In an experiment, the water of crystallisation is removed from hydrated Epsom salts to find x in the formula: **MgSO₄.xH₂O**
- The apparatus below is used to gather the information in the table.



(a) Mass of crucible	22,480	g
(b) Mass of crucible + Epsom salts	24,943	g
(c) Mass of crucible + Epsom salts - water	23,683	g
(d) Mass of Epsom salts	2,463	g
(e) Mass of Epsom salts - water	1,203	g
(f) Mass of water	1,260	g
(g) Moles of Epsom salts - water	4.3.1	mol
(h) Moles of water	4.3.2	mol
(i) ratio moles MgSO ₄ : moles H ₂ O	4.3.3	
(j) Formula of Epsom salts	4.3.4	

Complete the table by writing the question number in the block in your answer book. (4)

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QUESTION 7

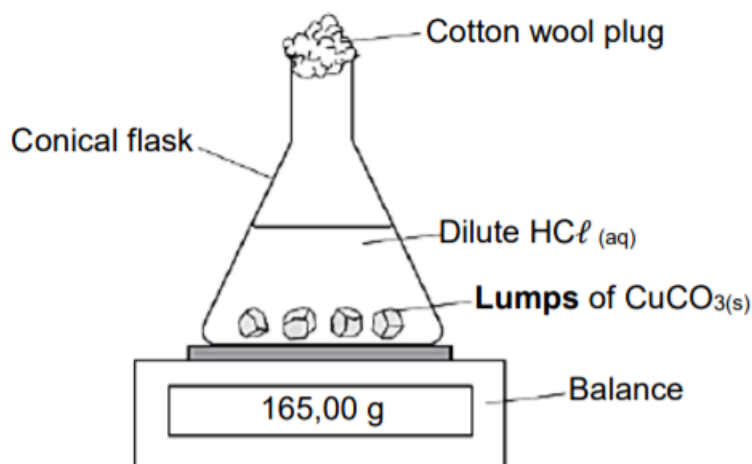
29 MARKS

Copper (II) carbonate reacts with excess dilute hydrochloric acid as shown.



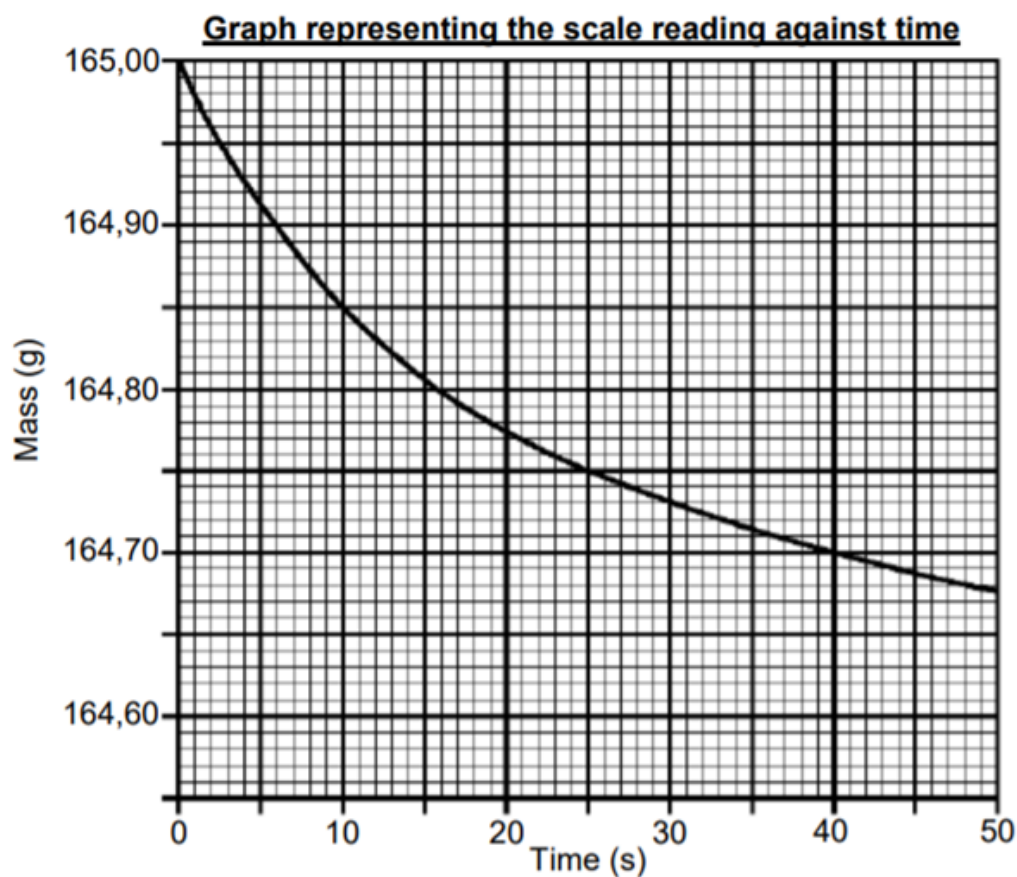
A student used the apparatus shown below to follow the progress of the reaction.

The copper (II)chloride solution produced has a green/blue colour and CO_2 is partially soluble in water.



The experiment was carried out using 0,5 g of copper (II) carbonate.

The graph below shows the results obtained for the first 50 s of the reaction.



- 7.1 Why is a cotton wool plug placed at the mouth of the flask? (2)
- 7.2 Define the *rate of the reaction*. (2)
- 7.3 Has the reaction gone to completion? Provide a suitable explanation for your answer. (2)
- 7.4 This is not the most ideal method to monitor this reaction. Suggest another method and explain how you would measure the rate of the reaction. (2)
- 7.5 Calculate the average rate of reaction in **mol.s⁻¹** between $t = 10\text{ s}$ and $t = 30\text{ s}$. (4)
- 7.6 The graph has been replicated on the answer sheet. On this graph sketch further relative graphs (no values required on the axes) if the following conditions have been changed. Label the graphs **a)** and **b)** accordingly.
- a) powdered copper (II) carbonate is used instead of the lumps. (2)
 - b) The concentration of the acid is decreased. (2)
- 7.7 In another experiment using the same chemicals, 1,5 g of impure copper (II) carbonate is reacted with 75 ml of 0,25 mol.dm⁻³ of hydrochloric acid to produce 180 cm³ of carbon dioxide gas at STP.
- 7.7.1 Calculate the percentage purity of the copper (II) carbonate. (6)
 - 7.7.2 Define *concentration*. (2)
 - 7.7.3 Determine the new concentration of the hydrochloric acid after the reaction is completed. (5)
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QUESTION 8 ORGANIC CHEMISTRY**24 MARKS**

Study the following compounds and answer the questions that follow.

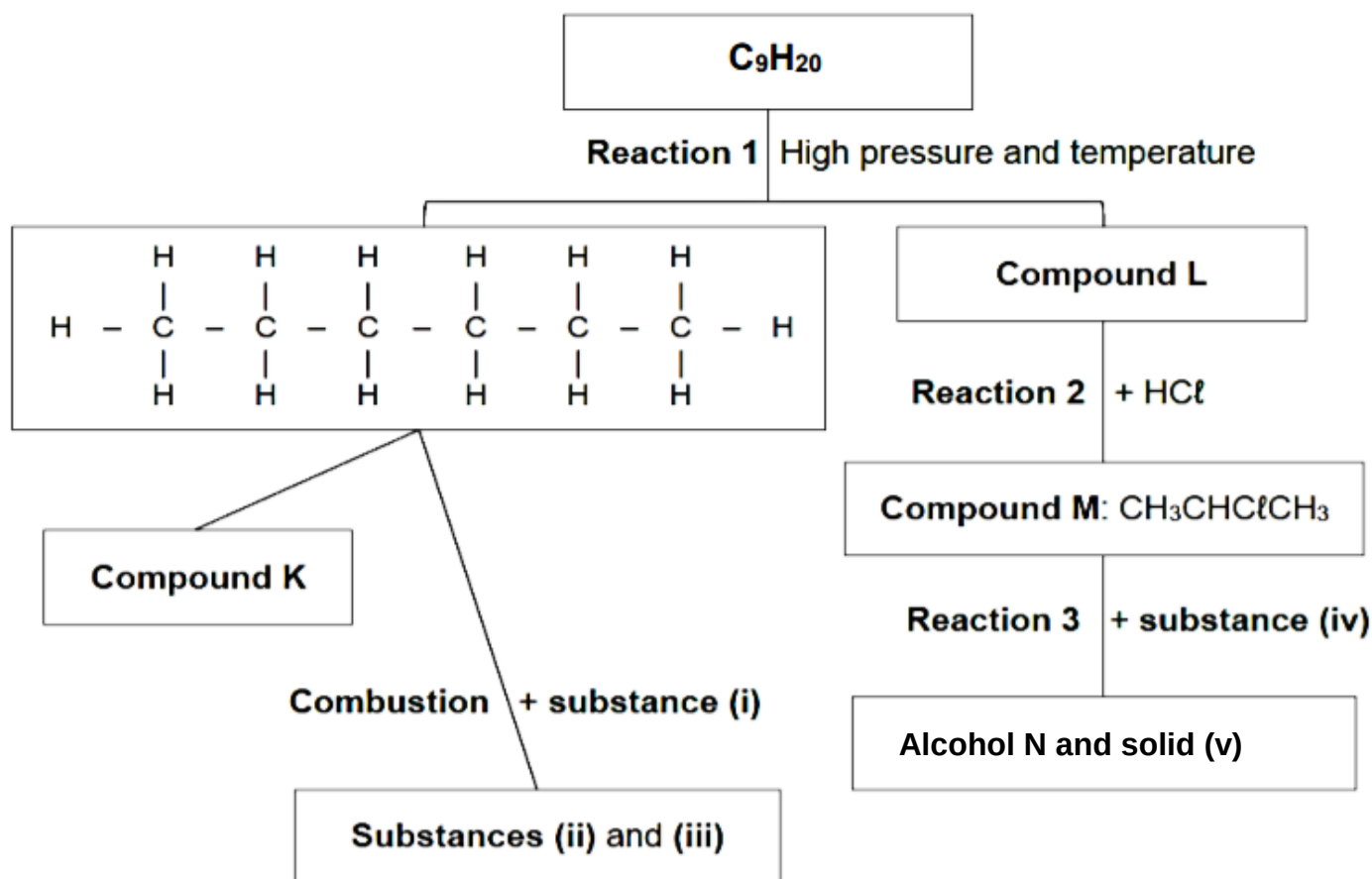
A. CH₃OH	D. CH₃CH₂COOH
B. CH₂CCHCH₂CH₃	E. HOCH₂CH₂CH₂OH
C. CH₃(CH₂)₄CH₃	F. CH₃CH₂CH(CH₃)CH₂CH₃

- 8.1 Give the IUPAC names for compounds **B** and **E**. (4)
- 8.2 Consider compounds **C** and **F**.
- 8.2.1 Define *isomer*. (2)
- 8.2.2 Which compound would have a higher boiling point (C or F)? (1)
- 8.2.3 Give a reason for your answer. (2)
- 8.3 Bromine water is added to **B** and **C** respectively.
- 8.3.1 Define *unsaturated compound*. (2)
- 8.3.2 Which compound (**B** or **C**) would rapidly decolour the bromine water? (2)
- 8.4 Compound **A** and **D** can undergo a condensation reaction under the correct conditions.
- 8.4.1 Using structural formula give the reaction equation between **A** and **D**. Include reaction conditions. (5)
- 8.4.2 Give the name AND molecular formula of the functional isomer of the organic product of this reaction. (2)
- 8.5 Define *homologous series*. (2)
- 8.6 Name the homologous series for compound **D**. (1)
- 8.7 Write down the general formula for compound **C**. (1)
-

QUESTION 9 ORGANIC CHEMISTRY

32 MARKS

A long chain hydrocarbon containing 9 carbons undergoes Reaction 1 to produce hexane and compound L. Further reactions of each substance are shown in the diagram below.



- 9.1 For **Reaction 1**, identify both the
- 9.1.1 GENERAL type of reaction and (1)
 - 9.1.2 SPECIFIC type of reaction that has occurred. (1)
- 9.2 Write the balanced combustion reaction for hexane hence identifying substances **(i)–(iii)**. Use molecular formulae.) (3)
- 9.3 Compound **K** is a structural isomer of hexane.
- 9.3.1 Give the IUPAC name for a possible isomer of hexane. (3)
 - 9.3.2 Write condensed structural formulae for compound K. (2)
- 9.4 Give the IUPAC name for Compound **L**. (2)
- 9.5 Refer to **Reaction 2**.
- 9.5.1 Identify the SPECIFIC type of reaction that took place. (2)
 - 9.5.2 What specific condition must be met for this reaction to take place? (2)

- 9.6 Refer to **Reaction 3** which produces an **alcohol** and a **solid**
- 9.6.1 Identify the GENERAL type of reaction that took place. (2)
- 9.6.2 Draw the structural formula for alcohol **N** and circle and name the functional group (3)
- 9.6.3 Identify substance **(iv)** and solid **(v)**. (2)
- 9.6.4 State the conditions required for this reaction to occur. (2)
- 9.7 Consider Compounds **L**, **M** and **N**.
- 9.7.1 Identify the type of intermolecular forces present in each of the compounds. (3)
- 9.7.2 Rank these compounds in increasing order of boiling points. (2)
- 9.7.3 Explain your choice of compound with the highest boiling point by referring to relative strengths of intermolecular forces. (2)
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END OF EXAMINATION